

Project 5 - Math 3250

March 22, 2026

1.a · It would cost \$37.00 total dollars to bet on all 36 total black/red squares plus 1 green square, guaranteeing you win on any spin outcome.

1.b · Every spin you would win back \$36.00, because of the 35:1 odds where you would win \$35.00 plus getting back your original \$1.00 bet.

1.c · $\text{profit} = \$36.00 - \37.00 so, $\text{profit} = -\$1.00$.

1.d ·

i · on any given spin of the wheel, if you bet \$1.00 on every slot then the profit is guaranteed to be \$35.00 and \$1.00 additionally from the winning slot; regardless of the outcome of the spin.

ii · on any given spin of the wheel, if you place a \$1.00 bet on every slot than your cost to play profit is -\$37.00.

iii · on any given spin of the wheel (no matter the outcome of the spin):

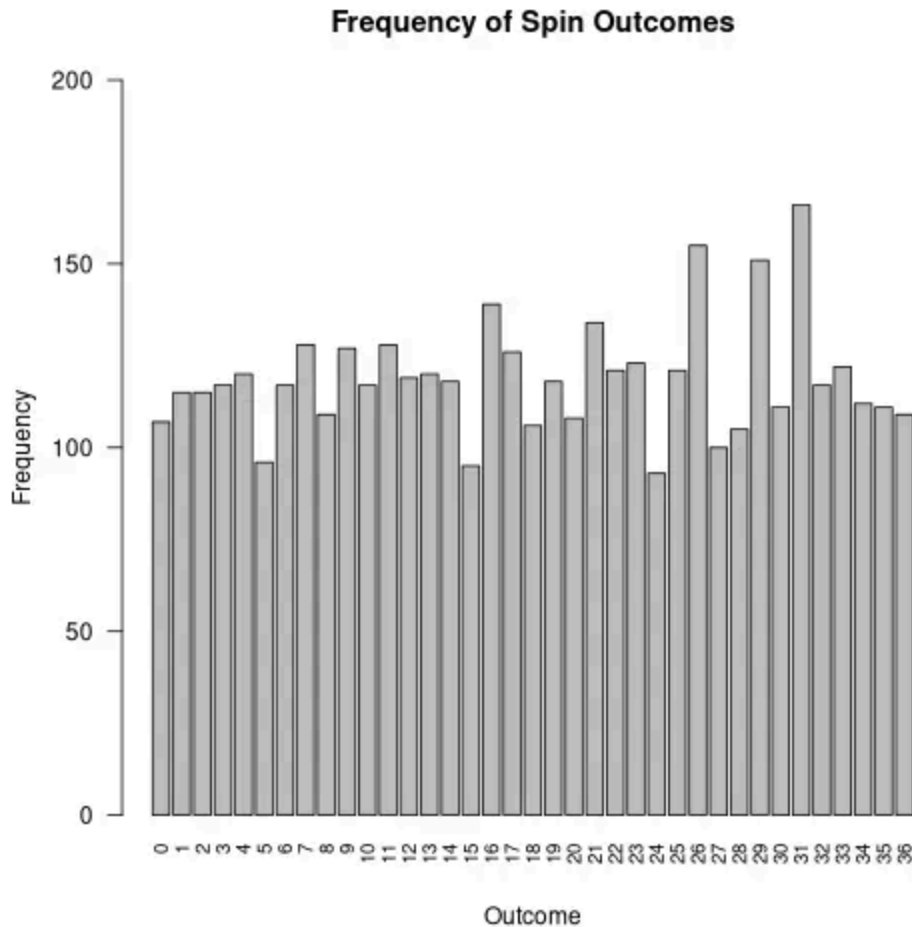
$$\text{total profit} = (\text{win mult} * \text{bet cost}) - (\text{bet cost} * \text{num bets}) + \text{bet cost}$$

$$\text{total profit} = (35 * 1) - (37 * 1) + 1$$

$$\text{total profit} = (35 - 37 + 1)$$

$$\text{total profit} = -1$$

2.a ·



2.b · The following values are the three spots where the spins landed the most frequently:

$$N_1 = 31, f = 166$$

$$N_2 = 26, f = 155$$

$$N_3 = 29, f = 151$$

2.c · The expected probability for any given square is $\frac{1}{37}$ or 0.02703 , so all three probabilities found for the highest frequency numbers are much higher than expected.

$$P(N_1) = 0.03776$$

$$P(N_2) = 0.03526$$

$$P(N_3) = 0.03435$$

3.x · Complete table with Fair and “Tisch” table probabilities under 3.e.

3.a · Given the information below, we can assume that it is extremely unlikely (0% chance when rounded) for a single square on the roulette table to be landed on over 6 times if you play 36 times total. Anything over 4 spins winning has less than a 1% chance. You have a $\approx 37\%$ chance of breaking even or losing 36 dollars and have $\approx 25.41\%$ of becoming profitable if you win at least 2 times. All this considering the wheel is perfectly fair.

3.b · You should lose money or break even $\approx 74.58\%$ of the times that you bet on 36 wheel spins.

3.c · You should become profitable only $\approx 25.42\%$ of the time if you when at least 2 spins or more of the 36.

3.d · Betting on the most frequent square on the “tisch 1” table, the odds of you losing or breaking even have gone down to ≈ 60.36 which makes your odds of making profit $\approx 39.64\%$.

3.e · The odds of making profit based on the calculated table of values is ≈ 0.3964 or $\approx 39.64\%$

Wins	Fair Probability	Tisch 1 Table Probability	Net Payout
0	0.3729	0.2502	-36
1	0.3729	0.3534	0
2	0.1813	0.2427	36
3	0.0571	0.1079	72
4	0.0131	0.0349	108
5	0.0023	0.0088	144
6	0.0003	0.0018	180
7	0.0000	0.0003	216
8	0.0000	0.0000	252
9	0.0000	0.0000	288
10	0.0000	0.0000	324
11	0.0000	0.0000	360

Wins	Fair Probability	Tisch 1 Table Probability	Net Payout
12	0.0000	0.0000	396
13	0.0000	0.0000	432
14	0.0000	0.0000	468
15	0.0000	0.0000	504
16	0.0000	0.0000	540
17	0.0000	0.0000	576
18	0.0000	0.0000	612
19	0.0000	0.0000	648
20	0.0000	0.0000	684
21	0.0000	0.0000	720
22	0.0000	0.0000	756
23	0.0000	0.0000	792
24	0.0000	0.0000	828
25	0.0000	0.0000	864
26	0.0000	0.0000	900
27	0.0000	0.0000	936
28	0.0000	0.0000	972
29	0.0000	0.0000	1008
30	0.0000	0.0000	1044
31	0.0000	0.0000	1080
32	0.0000	0.0000	1116
33	0.0000	0.0000	1152
34	0.0000	0.0000	1188
35	0.0000	0.0000	1224
36	0.0000	0.0000	1260